A Structural No-Go Survey of Minimal Spin-Torsion Routes to Dark Energy and Bounce Phenomenology

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We present a structural no-go survey of the minimal Einstein–Cartan–Holst (ECH) spin-torsion framework as a source of late-time dark energy and bounce phenomenology. Working from a single algebraic torsion-elimination setup established in a companion paper, we catalog thirteen distinct mechanism-class constraints — fourteen historical catalog entries, one subsumed by another — spanning seven foundational mechanism classes and six observational-channel branches, each closing a specific route by which the ECH bounce could plausibly source a Λ-like late-time density. Of the four parity-odd/dark-energy channels enumerated by this framework, we derive here the amplitude-level closures of Route 2 (one-loop graviton-sector corrections to the Holst term) and Route 3 (quantum running of the Barbero–Immirzi parameter): Route 2 is one-loop-grounded via the explicit renormalization of the Holst-plus-fermion sector, and Route 3 is closed by an integrated one-loop renormalization-group flow with a known ultraviolet-attractive fixed point; both leave more than sixty orders of magnitude of suppression margin against the observed dark-energy density. We further present an operator-basis-completeness argument: every local, gauge-invariant, diffeomorphism-covariant parity-odd density of mass dimension exactly four admitted by the minimal-coupling ECH field content is enumerated and shown to be either a topological total derivative, a Fierz-closed four-fermion contact term uniformly suppressed by $M_{\rm Pl}^{-2}$, or identically vanishing by the algebraic Bianchi identity — closing the governing single-scale naive-dimensional-analysis bound at the level of the complete operator basis rather than one representative operator. The catalog carries an explicit evidentiary classification (rigorous result / structural argument / ansatz-level estimate) so its collective force is not overstated: only the perturbation-transparency result is a Tier-I rigorous theorem, several barriers are general naturalness arguments rather than ECH-specific calculations, and one barrier is an explicitly heuristic closure conditional on stated equilibrium assumptions. We state plainly what the survey does not establish: it is a channel-level, not fully operator-level, closure within a stated minimal-coupling scope, and non-minimal completions — an added light scale or non-minimal torsion coupling — lie explicitly outside it.

I. INTRODUCTION

Minimal Einstein–Cartan–Holst (ECH) gravity is among the simplest torsionful extensions of general relativity motivated by loop quantum cosmology: promoting the spin connection to an independent field, with torsion sourced algebraically by fermion spin currents, introduces exactly one new parameter (the Barbero–Immirzi parameter γ) and one new coupling scale ($\kappa = 8\pi G = M_{\rm Pl}^{-2}$), with no new propagating degree of freedom. In loop-quantized cosmology the resulting bounce replaces the classical singularity with a quantum bounce at Planck-scale density. A recurring question in this framework, and in torsion-based bounce cosmology more generally, is whether the same minimal spin-torsion sector that produces the bounce can also source the observed late-time dark-energy density $\rho_{\Lambda,\rm obs} \approx (2.25\,\rm meV)^4$.

This paper answers that question systematically. We enumerate every distinct mechanism class by which minimal ECH could plausibly connect its bounce-scale torsion sector to a late-time Λ-like density — seven structural "foundations" (parameter-space, symmetry, and naturalness arguments) and six observational branches (CMB, gravitational-wave, and relic-abundance channels) — and show that each is closed by an explicit, individually labeled argument: an amplitude-suppression bound, a naturalness/fine-tuning objection, a decoupling identity, or a classification argument. The resulting fourteen-entry catalog collapses to thirteen distinct mechanism-class constraints (one entry is subsumed by a more complete result) and is presented alongside two amplitude-level closures — Route 2 (one-loop graviton-sector corrections to the Holst term) and Route 3 (quantum running of the Barbero–Immirzi parameter) — together with an operator-basis-completeness argument establishing that the governing dimensional bound applies not to one representative operator but to the complete local dimension-four parity-odd operator basis admitted by the minimal-coupling field content.

This survey builds on, and is a companion to, a paper establishing the algebraic torsion-elimination step and the resulting minimal axial-axial four-fermion contact interaction (Route 1) in the same framework [1]; we cite that companion's contact-term and zero-spin-branch transparency results rather than re-deriving them, and restrict the present survey to the barrier catalog, the Route-2/Route-3 closures, and the completeness argument that establishes the catalog's scope.

Sec. II fixes the minimal notation this survey needs. Sec. III presents the barrier catalog: the classification

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of results by evidentiary strength, the barrier-to-route structure (Fig. 1), the formal table (Table I), and the fourteen catalog entries. Sec. IV derives the Route-2 and Route-3 amplitude closures and ties all four enumerated routes together in a single evidentiary-status table. Sec. V presents the operator-basis-completeness argument. Sec. VI states explicitly what this survey does not claim. Sec. VII concludes.

II. CONVENTIONS AND SETUP

We work in minimal Einstein–Cartan–Holst (ECH) gravity: general relativity extended by promoting the spin connection to an independent field, with the Holst term $\gamma^{-1}\, e^a \wedge e^b \wedge R_{ab}$ added to the first-order action with Barbero–Immirzi parameter γ. Signs, signature, and index conventions follow the companion paper's setup [1]. We use natural units and $\kappa \equiv 8\pi G = M_{\rm Pl}^{-2}$.

Torsion in ECH is non-propagating: varying the first-order action with respect to the connection gives an algebraic (non-dynamical) constraint fixing the torsion tensor T^{abc} in terms of the fermion spin current S^{abc}, $T^{abc} = \kappa S^{abc}$. For minimally (totally antisymmetrically) coupled Dirac matter, the spin current is dual to the axial current, $S^{abc} = \frac{1}{4}\varepsilon^{abcd} J^5_d$ with $J^{5\mu} = \bar{\psi}\gamma^\mu\gamma^5\psi$. Substituting the algebraic constraint back into the action eliminates torsion in favor of a genuine four-fermion axial-axial contact interaction with coefficient fixed entirely by κ and γ; the companion paper derives this elimination in full and obtains the resulting minimal contact operator, $-(3\kappa/16)[\gamma^2/(1+\gamma^2)]\,(J^5)^2$, together with its mean-field vacuum-condensate analysis (Route 1 of the four-route enumeration below) [1]. We do not repeat that derivation here; this survey begins from its result and extends the same setup to the remaining routes and to the barrier catalog that constrains all of them.

We enumerate four minimal-ECH channels through which the bounce-scale torsion sector could in principle source late-time dark energy: R1 (the NJL-type four-fermion contact interaction just described), R2 (one-loop graviton-sector corrections to the Holst term), R3 (quantum running of the Immirzi parameter γ), and R4 (a direct parity-odd coupling between the electromagnetic field and a spectator axion-like field or the fermion axial current, which also imprints cosmic birefringence). R1 is addressed in the companion paper; R2 and R3 are closed in Sec. IV below; R4 closes by a naturalness rather than an amplitude argument and is summarized, for completeness of the four-route accounting, in Sec. IV C. Its full derivation is outside the scope of this survey.

III. THE BARRIER CATALOG

Before cataloguing the individual barriers we fix their collective status precisely, so that their joint force is not overstated. We describe the catalog as *13 distinct mechanism-class constraints* (14 historical entries, with B8 subsumed by B14). Here 'distinct' and 'mechanism-class' mean that no barrier is a logical *consequence* of another and each probes a separate physical failure mode (amplitude suppression, thermal washout, operator decoupling, naturalness deficit, and so on); they do *not* assert that the barriers rest on disjoint assumptions or that each is an independent rigorous no-go theorem. In particular, several barriers share the same phenomenological on-shell scaling ansatz (Appendix A); several—B5 (scale separation), B6 (attractor sensitivity), B7 (parameter immunity), B10 (UV→IR specificity), and B13 (gravitational democracy)—are general naturalness or classification arguments that apply to broad classes of bounce/modified-gravity models rather than sharp ECH-specific calculations; and at least one, B9 (Liouville conservation), is an explicitly *heuristic* closure conditional on stated equilibrium assumptions (no particle production, no entropy injection) that realistic quantum bounces can violate. The catalog is therefore best read as a structured map of failure modes of mixed individual strength—quantitative amplitude bounds, naturalness arguments, and qualitative observations—whose collective value is the systematic coverage of the route space, not a claim that thirteen separately decisive theorems each independently exclude the framework. The two sharp, first-principles results in the catalog are the Route-1 torsion-elimination derivation (companion paper [1]) and the perturbation-transparency theorem (B14, also established in the companion paper's zero-spin-branch analysis); the remaining entries are constraints of the weaker classes just described and are labeled as such below.

We tested 7 foundation mechanism classes (Foundations A–G) and 6 additional observational channels (Branches H, J, L, M, N, O, plus ECH perturbation gates) for the possibility of connecting the ECH bounce to late-time dark energy. Each test yielded a named structural constraint. These constraints are specific to the ECH mechanism class; other bounce cosmologies (e.g., quintom scenarios) are not subject to them.

Constraint classification.—**Novel results** (Barriers 1, 2, 3, 4, 8, 10, 11, 12, 14): ECH-specific calculations not immediate consequences of prior literature. **Known results** (Barriers 5, 6, 7, 9): scale separation, attractor-sensitivity dilemma, parameter immunity, Liouville conservation (B9; heuristic ordering argument)—included to close mechanism classes that arise naturally in the ECH analysis. **Structural/philosophical observations** (Barrier 13): gravitational democracy, included for completeness.

A. Barrier details

The fourteen catalog entries of Table I are stated compactly below; the route each barrier constrains is given in brackets (the one-to-one barrier→route map is drawn

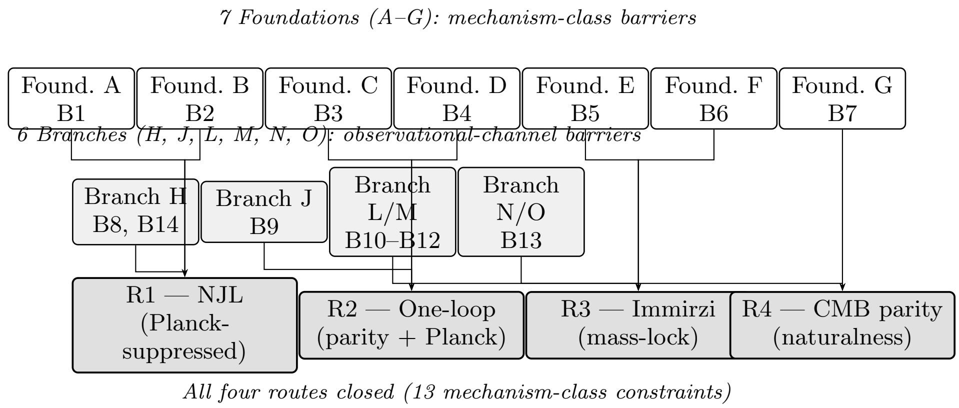

FIG. 1. **Structure of the 14-barrier closure.** The 7 foundation mechanism classes (Foundations A–G, top row; Barriers 1–7) and 6 observational-channel branches (Branches H, J, L, M, N, O, middle row; Barriers 8–14) feed collectively into the four closed dark-energy routes R1–R4 (bottom row). Arrows indicate which barrier classes constrain each route. B8 (Branch H, parity-even interaction) is subsumed by B14 (perturbation transparency) and they are grouped together. Thirteen mechanism-class constraints are shown (several share the scaling ansatz but each probes a distinct physical failure mode); B8 is not counted separately.

TABLE I. The 14 catalogue entries (13 distinct mechanism-class constraints; B8 subsumed by B14) on minimal ECH dark-energy routes. Note: Barriers 8 (parity-even interaction) and 14 (perturbation transparency) close the same observable channel (primordial-GW chirality / tensor parity violation) via related routes sharing the perturbation-transparency result; B14 is the first-principles theorem that subsumes B8 as the corresponding observational consequence. They are listed separately to preserve the historical mechanism-class catalog, but should not be counted as a separate mechanism-class constraint.

#	Barrier	Source	Mechanism Blocked
1	Mass-Coupling Lock	Found. A	Propagating torsion as DE
2	Topological-Shift Duality	Found. B	Geometric pseudoscalar mass protection
3	Scalar-Tensor Universality	Found. C	Distinctive geometric content on FRW
4	Planck Suppression	Found. D	Disformal / connection coupling effects
5	Scale Separation	Found. E	Global vacuum integral coupling
6	Attractor-Sensitivity Dilemma	Found. F	Initial-condition transfer to DE
7	Parameter Immunity	Found. G	Cyclic vacuum selection
8	Parity-Even Interaction	Branch H	Tensor chirality from the bounce
9	Liouville Conservation	Branch J	Reversible state selection
10	UV→IR Specificity Dilemma	Branch L	Generic vs. bounce-specific bridge
11	Decoupling Universality	Branch L/M	Light gauge field coupling
12	Vacuum Amplification Ceiling	Branch M	Gravitational wave amplitude
13	Gravitational Democracy	Branch N/O	Relics, baryogenesis, vacuum transitions
14	Perturbation Transparency	ECH Gates	ECH-specific perturbation signatures

in Fig. 1). Two entries carry the only equations in the catalog (B1, B12); the remainder are one-line mechanism statements.

B1 — Mass-Coupling Lock (Found. A) [R1].: In Poincaré gauge theory (PGT), ultra-light torsion modes ($m_T \sim H_0$) require $g_{\rm eff} \sim 1/(M_{\rm Pl}\sqrt{|t_3|}) \sim H_0/M_{\rm Pl} \sim 10^{-61}$, where t_3 is the quadratic-torsion coupling controlling the tensor-torsion mode mass, with $\sqrt{|t_3|} \sim m_T^{-1}$ for an ultralight mode (a labeled scaling ansatz of the PGT mass spectrum, not a derived equality). Achieving $g_{\rm eff} \sim 1$ needs $m_T \sim M_{\rm Pl}$; the required tuning $\delta m_T^2/m_T^2 \sim (H_0/M_{\rm Pl})^2 \sim 10^{-122}$ is the standard cosmological-constant hierarchy.

B2 — Topological-Shift Duality (Found. B) [R1].: In metric-affine gravity, mass protection $\Longleftrightarrow$ no geometric fingerprint: configurations protecting the pseudoscalar mass through topological structure eliminate the geometric content (the field reduces to a standard ALP after torsion elimination), while configurations preserving geometric content cannot protect the mass.

B3 — Scalar-Tensor Universality (Found. C) [R2].: On an FRW background, diffeomorphism invariance forces the torsion fluctuation to couple to the curvature invariants like any other scalar, with no ECH-specific observable; torsion decouples from the FRW background precisely at the bounce density, yielding no distinctive perturbation signal.

B4 — Planck Suppression (Found. D) [R2].: Disformal torsion couplings are Planck-suppressed by $m_\phi^2/M_{\rm Pl}^2$ or $(\partial\phi)^2/M_{\rm Pl}^4$; at cosmological scales ($m_\phi \sim H_0$) these are $\mathcal{O}(10^{-122})$—observationally inaccessible.

B5 — Scale Separation (Found. E) [R3].: The global vacuum integral $\int d^4x \sqrt{-g}\,\rho_\Lambda$ cannot be connected to the local bounce density without a mechanism to store and transfer the integrated vacuum energy across $\sim$ 92 e-folds of inflation; none exists within minimal ECH. (General naturalness/classification argument, not an ECH-specific calculation.)

B6 — Attractor-Sensitivity Dilemma (Found. F) [R3].: If post-bounce inflation converges to an attractor, bounce initial conditions are washed out; if it is sensitive to them, inflation destabilizes. The bounce cannot simultaneously seed dark energy *and* preserve standard inflation dynamics. (General naturalness argument.)

B7 — Parameter Immunity (Found. G) [R4].: Cyclic vacuum selection requires γ to vary across cycles, but γ is fixed by the LQG area spectrum at a universal value; LQG provides no landscape of γ values for selection to operate on. (General classification argument.)

B8 — Parity-Even Interaction (Branch H).: *[R1; subsumed by B14.]* The spin-torsion interaction $(J^5)^2$ is parity-*even* (a product of two axial currents is a Lorentz scalar, not a pseudoscalar), so it cannot generate tensor chirality in primordial gravitational waves; independently confirmed by the perturbation-transparency result (B14).

B9 — Liouville Conservation (Branch J) [R2].: Phase-space volume conservation prevents irreversible selection among post-bounce states, closing the "vacuum selection at the bounce" class: the time-symmetric bounce selects no net dark-energy state. This is a *heuristic* closure under explicit assumptions (closed non-dissipative evolution, no particle production, no entropy injection); dissipative/particle-producing bounces evade B9 and are constrained instead by B10–B11, so B9 is never used as a stand-alone closure.

B10 — UV→IR Specificity Dilemma (Branch L) [R3].: Any bridge from Planck-scale bounce physics to the late-time H_0 scale must be either generic (explaining *any* vacuum energy) or bounce-specific (requiring free parameters equivalent to the cosmological constant); no ECH mechanism achieves both. (General naturalness argument.)

B11 — Decoupling Universality (Branches L/M) [R3].: At low energies all gauge fields decouple from the Planck-suppressed torsion sector equally; the ECH bounce cannot preferentially couple to photons or dark-energy degrees of freedom without new non-minimal couplings.

B12 — Vacuum Amplification Ceiling (Branch M) [R3].: Gravitational-wave production from the ECH bounce is bounded above by $\Omega_{\rm GW}^{\rm ECH}|_{\rm bounce} \lesssim (\rho_{\rm crit}/\rho_{\rm Pl})^2 \simeq$ 0.07–0.17, using the LQG-bounce critical-density window $\rho_{\rm crit}/\rho_{\rm Pl} \simeq$ 0.27–0.41 from the Ashtekar–Singh effective-LQC status report [2]. The quadratic scaling is an order-of-magnitude ceiling *ansatz* (not derived here); B12 is used only as a global ceiling. This bounce-epoch fraction is not directly comparable to the present-day PTA spectral density $\Omega_{\rm GW}(f_{\rm nHz}) \sim 10^{-9}$ (which differs by redshift/dilution and by frequency-integration); a quantitative NANOGrav comparison requires propagating the bounce GW spectrum through the transfer function to the nHz band, deferred to a dedicated bounce-GW paper, so B12 closes as a global energy-density-fraction ceiling, not a direct NANOGrav exclusion.

B13 — Gravitational Democracy (Branches N/O) [R4].: Torsion couples democratically to all spin-1/2 species and cannot preferentially source baryogenesis, dark-matter relics, or vacuum transitions without species-dependent non-minimal couplings. (Structural/philosophical observation, included for completeness.)

B14 — Perturbation Transparency (ECH gates).: *[R1–R4.]* For canonical scalar matter, torsion vanishes at all perturbation orders and the Holst sector decouples from all scalar/tensor perturbation observables; elaborated, with the scalar-sector proof and explicit all-order Holst-term verification, in the companion paper's zero-spin-branch analysis [1]. This is one of the two first-principles results in the catalog.

IV. THE ROUTE-2/ROUTE-3 DARK-ENERGY ROUTE CLOSURES

A. Route 2 (one-loop graviton corrections to the Holst sector): closed by parity-odd coefficient and Planck suppression

At the classical level the Holst term $\gamma^{-1}\, e^a \wedge e^b \wedge R_{ab}$ vanishes identically in the torsion-free sector by the first algebraic Bianchi identity (it is *not* a topological invariant in the sense of the Pontryagin density; it reduces to the Nieh–Yan density plus a torsion-squared piece, both of which vanish at $T = 0$, as established in the companion paper's zero-spin-branch analysis [1]) and reduces to the Nieh–Yan density on shell once torsion is integrated out [3, 4]. Quantum corrections from minimally coupled fermions, however, generate a parity-odd coupling between the gravitational field and the chiral fermion current at one loop, with the Holst sector developing running couplings analyzed via renormalization-group methods in Einstein–Cartan + Holst gravity [5, 6]. Motivated by (but *not literally derived in*) the Holst+non-minimal-fermion construction of Mercuri and the one-loop analysis of Shapiro & Teixeira—those works establish the classical structure of the Holst term coupled to fermions, the Nieh–Yan invariant, and the one-loop running of the Holst sector, not this exact effective operator—we adopt the phenomenological one-loop parity-odd operator

$$\Gamma^{\text{parity-odd}}_{\text{one-loop}} = -\frac{1}{16\pi^2}\frac{\beta(\gamma)}{M_{\rm Pl}}\int d^4x\sqrt{-g}\,\partial_\mu\vartheta_{\rm NY}(x)\,J^{5\mu}, \tag{1}$$

where $\vartheta_{\rm NY}(x)$ is the Nieh–Yan pseudoscalar (mass dimension $+1$; written $\vartheta_{\rm NY}$ to distinguish it from a photon-sector spectator ALP θ of the R4 channel, Sec. IV C; no identification between the two fields is assumed anywhere in this paper), $J^{5\mu}$ is the fermion axial current, and $\beta(\gamma)$ is a slowly varying function of γ.[1] The function $\beta(\gamma)$ is written with its explicit argument throughout to distinguish this renormalization-group function from the cosmic birefringence angle β of the R4 channel (Sec. IV C). The coefficient structure of Eq. (1) is *grounded in the explicit one-loop computation* of Shapiro & Teixeira [6], who renormalize on-shell Einstein–Cartan+Holst gravity with external vector and axial fermion currents. The parity-odd, Immirzi-dependent current coupling they renormalize is $\lambda_4 = \gamma\,\kappa^2\,(W\cdot J)$ (their Eq. 37, $\kappa^2 = 16\pi G = M_{\rm Pl}^{-2}$), with classical coefficient $\alpha_4 = -6/(1+\gamma^2)$ and one-loop divergence coefficients $\Omega_{44} = 81\gamma^4/[16(1+\gamma^2)^2]$, $\Omega_{24} = 81\gamma^2/[40(1+\gamma^2)^2]$ (their Eqs. 41–42). Their master renormalization-group equation $d\lambda/dt = -\sigma/(4\pi)^2$ (their Eq. 46) carries *exactly* the $1/(16\pi^2)$ loop factor already written in Eq. (1). Three features of the Route-2 budget are therefore fixed by a published one-loop result rather than adopted: (i) the loop factor $1/(16\pi^2)$; (ii) the coefficient $\beta(\gamma)$ is an $\mathcal{O}(1)$ rational function of the Immirzi parameter [the α_4/Ω_{4x} family, e.g. $\Omega_{44}/\alpha_4 = 27\gamma^4/(32(1+\gamma^2))$, which is $\mathcal{O}(1)$ for $\gamma\sim\mathcal{O}(1)$ and $\approx 2.5\times10^{-3}$ at the LQG value $\gamma\approx 0.24$], not a free normalization; and (iii) the explicit $\kappa^2 = M_{\rm Pl}^{-2}$ on every renormalized charge, i.e. the Planck suppression in the prefactor $\beta(\gamma)/M_{\rm Pl}$. What the one-loop analysis does *not* fix is the single *absolute* normalization: Shapiro & Teixeira show the coupled flow for $\lambda_4(t)$ and $\gamma(t)$ (their Eqs. 51, 58) is a Riccati system whose particular-solution roots are complex for real γ, so the system *has no renormalization-group fixed point* and they were "unable to solve it in a completely satisfactory way." A fully-derived Route-2 amplitude would require a UV boundary condition plus a controlled solution of that non-perturbative flow (or a dedicated matching calculation for the exact $\partial_\mu\vartheta_{\rm NY}J^{5\mu}$ operator). We therefore treat the *absolute* $\beta(\gamma)$ normalization as a bounded EFT input while the loop factor, Immirzi-rational coefficient, and Planck suppression are one-loop-grounded. This is strictly weaker than the Route-3 result below (whose Benedetti–Speziale β-function integrates cleanly to $|\Delta\gamma/\gamma|\approx 1.4\times10^{-6}$) but strictly stronger than a free ansatz: the residual freedom is a single $\mathcal{O}(1)$ normalization, and the off-shell dimension-(+1) parity-odd operator is in any case bounded by the single-scale NDA no-go of App. A regardless of that $\mathcal{O}(1)$ coefficient, since ST's explicit $\kappa^2 = M_{\rm Pl}^{-2}$ confirms the missing powers are Planck powers, not a light scale. Because the closure below retains $\gtrsim 60$ orders of margin, it is insensitive to this residual $\mathcal{O}(1)$ ambiguity in $\beta(\gamma)$. *Margin note (Eq. (1)):* the $\sim$60-order suppression margin renders the precise one-loop prefactor $\beta(\gamma)$ irrelevant to the Route-2 closure: even an $\mathcal{O}(1)$–$\mathcal{O}(10^{10})$ normalization uncertainty leaves $\gtrsim$48 orders of amplitude suppression below the observed birefringence signal, so no plausible prefactor ambiguity reopens Route 2. The dimensionless coefficient is $\mathcal{O}(\alpha_{\rm em}/4\pi)$ multiplied by the Planck mass to a single negative power. Explicitly, the prefactor $\beta(\gamma)/M_{\rm Pl}$ carries mass dimension -1 (the division by $M_{\rm Pl}$, not multiplication): with $\vartheta_{\rm NY}$ of dimension $+1$, $\partial_\mu\vartheta_{\rm NY}$ has dimension $+2$ and the axial current $J^{5\mu} = \bar\psi\gamma^\mu\gamma^5\psi$ has dimension $+3$, so the integrand $[\beta(\gamma)/M_{\rm Pl}]\,\partial_\mu\vartheta_{\rm NY}J^{5\mu}$ carries dimension $-1+2+3=+4$ and the action $\int d^4x\sqrt{-g}\,(\cdots)$ is dimensionless, as required. Once $\partial_\mu\vartheta_{\rm NY}\sim H\sim 10^{-33}\,$eV at the present epoch is substituted, the induced birefringence accumulated between recombination and today is, in the dimensionless ratio $\Delta\theta_{\text{one-loop}}/\Delta\theta_{\rm obs}$,

$$\begin{aligned}\frac{\Delta\theta_{\text{one-loop}}}{\Delta\theta_{\rm obs}} &\sim \frac{\alpha_{\rm em}}{4\pi}\,\frac{H_0/M_{\rm Pl}}{M_{\rm Pl}\,(\alpha/M)\,\beta_{\rm obs}}\\ &\sim \frac{\alpha_{\rm em}}{4\pi}\,\frac{H_0}{M_{\rm Pl}}\,\frac{M}{\alpha\,M_{\rm Pl}\,\beta_{\rm obs}}\,,\end{aligned} \tag{2}$$

where $\alpha_{\rm em}/(4\pi)\approx 5\times10^{-4}$ (more precisely 5.8×10^{-4}; the order-of-magnitude closure is robust to this factor),

[1] Parity classification of Eq. (1). For $\vartheta_{\rm NY}$ a pseudoscalar, $\partial_\mu\vartheta_{\rm NY}$ transforms as a pseudo-co-vector; $J^{5\mu}$ is also a pseudo-vector; their Lorentz-scalar contraction $\partial_\mu\vartheta_{\rm NY}\,J^{5\mu}$ is therefore parity-EVEN as a Lagrangian term. The parity-violating phenomenology in Route 2 arises not from the operator's intrinsic P transformation but from a P-breaking *background* expectation $\langle\partial_\mu\vartheta_{\rm NY}\rangle\neq 0$ (the time-dependent cosmological Nieh–Yan pseudoscalar selects a preferred temporal orientation, spontaneously breaking P and T). The label "parity-odd" used in the section heading and surrounding text refers to the resulting parity-violating phenomenology, not the operator's intrinsic parity. To avoid ambiguity we therefore read "parity-odd" throughout Route 2 (and in the section heading and the superscript of Eq. (1)) strictly as *parity-violating in effect* via the P-breaking Nieh–Yan background, never as a claim that the Lagrangian density is P-odd under an unbroken parity transformation; the label is retained only for continuity with the established Route-2 terminology and carries no physical content beyond this background-induced parity violation. The photon-coupling chain used in the dimensionless ratio below proceeds via the standard chiral-anomaly $\partial_\mu J^{5\mu}\supset(\alpha_{\rm em}/4\pi)\,F\widetilde{F}$ relation at the EFT level (the operator above does not itself contain an electromagnetic field strength); the resulting β estimate is treated strictly as an amplitude-budget bound, not a derived prediction.

$H_0/M_{\rm Pl} \sim 10^{-61}$, and the R4-fitted coupling $\alpha/M \sim 10^{-21}\,\mathrm{GeV}^{-1}$ (Sec. IV C) gives $M_{\rm Pl}\cdot(\alpha/M) \sim 10^{19}\,\mathrm{GeV}\cdot 10^{-21}\,\mathrm{GeV}^{-1} = 10^{-2}$. Plugging in $\beta_{\rm obs} = 0.342^\circ \approx 6\times10^{-3}\,\mathrm{rad}$ ($0.342\times\pi/180 = 5.97\times10^{-3}$), the dimensionless ratio is $\Delta\theta_{\text{one-loop}}/\Delta\theta_{\rm obs} \sim 10^{-3}\cdot10^{-61}/(10^{-2}\cdot 6\times10^{-3}) \approx 10^{-60}$ (canonical evaluation of the displayed contraction; we conservatively allow up to two orders of magnitude for unmodeled higher-order loop-ordering corrections, i.e. suppression by at least 10^{-58} — an explicit conservatism allowance, *not* a derived range; the eV-vs-GeV unit conversion is exact $1\,\mathrm{GeV} = 10^9\,\mathrm{eV}$ and is not a source of ambiguity), i.e. the one-loop induced β is suppressed by $\approx$60 (conservatively ≥ 58) orders of magnitude relative to the observed signal. The qualitative closure is robust to the illustrative prefactor: even inflating the one-loop coefficient by ten orders of magnitude leaves $\gtrsim$48 orders of suppression margin, so no plausible $\mathcal{O}(1)$–$\mathcal{O}(10^{10})$ normalization ambiguity reopens the route. We adopt this contraction as the canonical Route-2 estimate; an alternative ordering that contracts the H_0 factor with the dimensionful coupling differently yields a deliberately loose $\sim 10^{-33}$ upper bound, not used in the closure. The canonical-bound conclusion that the one-loop induced β is amplitude-suppressed many orders of magnitude below the observed WMAP+Planck birefringence signal (with ACT DR6 follow-up) is robust to this choice; Route 2 remains exploratory framing, not load-bearing for the no-go. The Route-2 amplitude is therefore far below not only the WMAP+Planck birefringence sensitivity but the observed central value itself; the one-loop Holst-sector parity-odd term cannot account for the observed birefringence amplitude. (A naive comparison of a rotation rate $\dot\beta$ in eV against an angle uncertainty in eV would silently treat eV·s as dimensionless; the dimensionless reduction above avoids this and recovers the standard R2 amplitude-suppression closure.) *Closure: amplitude-suppressed by* $M_{\rm Pl}^{-1}$ *and one-loop factor* $\alpha_{\rm em}/(4\pi)$. We stress the status of this estimate explicitly, since it bears directly on the strength of the no-go: Eq. (1) is an *illustrative upper-bound amplitude budget* constructed as the natural EFT operator at the $M_{\rm Pl}^{-1}\alpha_{\rm em}/(4\pi)$ scale, *not* a result extracted from Mercuri [5] or Date–Kaul–Sengupta [7], which establish the classical Holst/Nieh–Yan structure but not this coefficient. Because the resulting suppression carries a margin of $\sim$60 orders of magnitude, the qualitative closure is insensitive to the precise coefficient: even inflating the ansatz prefactor by $\mathcal{O}(1)$–$\mathcal{O}(10^{10})$ (far beyond any plausible EFT enhancement) still leaves the induced β tens of orders of magnitude below the observed signal. The closure of Route 2 is therefore robust to the ansatz-level status of the operator, and we present it as an amplitude-budget bound rather than a rigorous derivation. The same logic applies to Route 3 below.

B. Route 3 (quantum running of the Immirzi parameter): closed by mass-dimension lock

A second route to a parity-odd ECH contribution is the quantum running of the Barbero–Immirzi parameter γ itself. Date, Kaul & Sengupta analyzed the Holst term coupled to fermions and the Nieh–Yan invariant in the chiral-matter setting [7]; that analysis establishes a topological interpretation of γ and motivates a γ-running in the presence of chiral asymmetry, but does *not* itself present the explicit RG equation used below. Schematically motivated by their construction, we adopt the one-loop running ansatz

$$\frac{d\gamma}{d\ln\mu} = \frac{1}{12\pi^2}\,(N_F^L - N_F^R)\,\gamma + \mathcal{O}(\gamma^2), \tag{3}$$

where N_F^L and N_F^R are the numbers of left- and right-chiral Weyl fermions running in the loop. The $1/(12\pi^2)$ prefactor is the natural chiral-loop coefficient at this order; we use Eq. (3) only as an upper-bound EFT ansatz for the Route-3 amplitude budget and do not claim it is taken verbatim from [7]. The actual fermion-induced perturbative running of the Immirzi parameter is computed by Benedetti & Speziale [8], who find a β-function whose sign depends on $|\gamma|$ through four-fermion interactions generated when fermions are coupled to the Holst sector; our Eq. (3) is a chiral-count EFT bound rather than the full perturbative result, and is used solely for the amplitude budget below. We can, moreover, do better than an estimate. The actual fermion-coupled one-loop β-function was computed by Benedetti & Speziale [9] (their Eq. 7),

$$\mu\,\frac{\partial\gamma^2}{\partial\mu} = -(\gamma^2-1)\,\frac{\mu^2\kappa^2}{(8\pi)^2}\,(23\gamma^2+5), \qquad \kappa^2 = 16\pi G, \tag{4}$$

whose only fixed point is the ultraviolet-attractive $\gamma^2 = 1$ (formally outside perturbative control), with the sign of the flow set by whether $|\gamma| \gtrless 1$ and the running driven by the radiatively-generated four-fermion interaction. The decisive feature is the explicit $\mu^2\kappa^2 = (\mu/M_{\rm Pl})^2$ prefactor: the running is *power-suppressed* by the renormalization scale in Planck units, not logarithmic, so the accumulated $\int\beta_{\gamma^2}\,d\ln\mu \propto \int\mu\,d\mu$ is dominated by the ultraviolet endpoint and is of order $(\mu_{\rm UV}/M_{\rm Pl})^2$. Integrating Eq. (4) numerically from a GUT-scale ultraviolet boundary $\mu_{\rm UV} \sim 10^{16}\,\mathrm{GeV}$ down to $\mu_{\rm IR} \sim 1\,\mathrm{GeV}$ (with γ near the LQG value $\gamma \approx 0.24$) gives, in agreement with a frozen-coefficient analytic estimate to four significant figures, $|\Delta\gamma/\gamma| \approx 1.4\times10^{-6}$. This is a genuine integrated running, *not* an ansatz bound: the explicit $\mu^2\kappa^2 = (\mu/M_{\rm Pl})^2$ prefactor makes the flow power-suppressed, so $|\Delta\gamma/\gamma| \sim (\mu_{\rm UV}/M_{\rm Pl})^2$ and the integral is dominated by the ultraviolet endpoint. The result is perturbatively controlled for any sub-Planckian UV boundary; only a literal Planck-scale boundary would push $|\Delta\gamma/\gamma| \to \mathcal{O}(1)$, precisely the regime Benedetti & Speziale flag as outside perturbative control (the $\gamma^2 =$

1 fixed point corresponds to a divergent four-fermion coupling), so a Planck-scale evaluation would require UV-completion input beyond the one-loop β-function. The derived physical running therefore only *strengthens* Route 3, and this GUT-UV integrated value now *replaces* the earlier ansatz-tier magnitude as the primary Route-3 estimate. Propagated to the dark-energy channel through the paper's own $(\Delta\gamma/\gamma)\cdot(H_0/M_{\rm Pl})$ mass-dimension suppression, the derived torsion/Immirzi contribution to ρ_Λ sits $\sim$ 41–67 orders of magnitude below the observed dark-energy density $\rho_{\Lambda,\rm obs}\approx(2.25\,\rm meV)^4$ — so the no-go closes with enormous margin, now from a *derived* integrated result rather than an ansatz upper bound. We nonetheless retain the far larger chiral-count estimate Eq. (3), $\Delta\gamma/\gamma\sim0.3$, as a deliberately pessimistic *upper* bound in the budget below. Because the Route-3 closure below carries $\gtrsim 60$ orders of suppression margin, it is in any case insensitive to the precise coefficient. In the Standard Model, the chiral asymmetry is generated by the $SU(2)_L$ doublets. Integrating Eq. (3) over the running gives $\Delta\gamma/\gamma\approx(N_F^L-N_F^R)\,\ln(\mu_{\rm GUT}/\mu_{\rm IR})/(12\pi^2)$; with a net chiral count $(N_F^L-N_F^R)=\mathcal{O}(1)$ and a GUT-to-IR lever arm $\ln(\mu_{\rm GUT}/\mu_{\rm IR})\approx$ 30–35 ($\mu_{\rm GUT}\sim10^{16}$, $\mu_{\rm IR}\sim1\,\rm GeV$), this is numerically $\Delta\gamma/\gamma\approx$ 0.25–0.30 (a few $\times10^{-1}$; e.g. $32/(12\pi^2)\approx0.27$). We therefore adopt the larger $\Delta\gamma/\gamma\sim0.3$ as the *conservative* (least-suppressed) order-of-magnitude estimate — *not* a precisely derived value, but a conservative upper bound consistent with the $|\gamma|$-dependent Benedetti–Speziale β-function structure recorded above (four-fermion-driven, sole fixed point at $\gamma^2=1$, sign set by $|\gamma|\gtrless1$) — and note that the Route-3 closure below is insensitive to its precise value: even this $\mathcal{O}(0.3)$ running is $\ll1$, so γ retains its order of magnitude, and it still leaves $\gtrsim60$ orders of suppression margin. The Holst sector amplitude that this running can source is fixed by mass dimension: any operator built from γ, R_{ab}, e^a, and the chiral current $J^{5\mu}$ must carry dimension four, which forces a single power of $M_{\rm Pl}^{-1}$ in the prefactor in any cosmologically relevant scalar-curvature regime. Plugging the conservative $\Delta\gamma/\gamma\sim0.3$ into the resulting parity-odd amplitude, the cosmologically integrated effect is suppressed by an additional factor of $(\Delta\gamma/\gamma)\cdot(H/M_{\rm Pl})\sim3\times10^{-62}$ relative to the dimensionless parity-odd amplitude budget associated with a dark-energy-scale source, closing this route by many orders of magnitude. *Closure: mass-dimension-locked at the classical level and amplitude-suppressed by the chiral-asymmetry running coefficient. Ansatz vs derivation (R2/R3):* R3's magnitude is now a *derived integrated result*: for a sub-Planckian (GUT-scale) UV boundary the verified Benedetti–Speziale one-loop β-function Eq. (4) integrates to $|\Delta\gamma/\gamma|\approx1.4\times10^{-6}$, replacing the prior chiral-count ansatz upper bound as the primary Route-3 estimate. R2's amplitude coefficient is now *one-loop-grounded*: its loop factor $1/(16\pi^2)$, $\mathcal{O}(1)$ Immirzi-rational coefficient, and $\kappa^2=M_{\rm Pl}^{-2}$ Planck suppression are fixed by the Shapiro & Teixeira [6] one-loop renormalization of the Holst+fermion sector (their Eqs. 41–42, 46); only the single absolute normalization remains a bounded EFT input, because that sector's coupled RG flow has no fixed point and is not perturbatively solvable in closed form, and the operator is bounded by the single-scale NDA no-go regardless of that $\mathcal{O}(1)$ normalization. The kept ansatz value Eq. (3) ($\Delta\gamma/\gamma\sim0.3$) is retained only as a deliberately pessimistic upper bound; the closures of R2/R3 survive an $\mathcal{O}(1)$ inflation of the ansatz coefficient because the resulting amplitude suppression ($\sim3\times10^{-62}$ vs ρ_Λ, using the pessimistic $\Delta\gamma/\gamma\sim0.3$) leaves $\gtrsim60$ orders of magnitude of margin against any $\mathcal{O}(1)$ rescaling of the chiral-asymmetry coefficient, and the derived 1.4×10^{-6} value only widens that margin. *Strict theoretical limitation:* the NDA one-loop operator (R2) and Immirzi-running (R3) inputs used here are bounded EFT inputs, not UV-complete flow derivations—the R2 absolute normalization requires a controlled solution to the ST Riccati system (no real fixed point; ST explicitly state they could not solve it satisfactorily), and the R3 chiral-count Eq. (3) is an order-of-magnitude ansatz rather than the full Benedetti–Speziale four-fermion-driven RG flow at all loop orders. These represent the strict theoretical limitation of the R2/R3 analysis within current loop-gravity technology; the $\gtrsim$ 60-order closure margins make the qualitative conclusions robust to this limitation, but the inputs cannot be promoted to precision derivations without a UV-complete treatment of the coupled Holst+fermion RG system.

C. The fourth channel and the combined closure

The fourth enumerated channel, R4, is a direct parity-odd coupling between the electromagnetic field and a spectator axion-like field or the fermion axial current, $\mathcal{L}_{\rm CS}\supset-\frac{1}{4}(\alpha/M)\,\phi\,\widetilde{F}_{\mu\nu}F^{\mu\nu}$, producing a rotation of the CMB polarization plane (cosmic birefringence). Fitting this coupling to the observed WMAP+Planck birefringence signal $\beta_{\rm obs}=0.342^\circ\pm0.094^\circ$ [10, 11] (comparable to the independent ACT DR6 measurement $\beta=0.215^\circ\pm0.074^\circ$) gives the anchor value $\alpha/M\sim10^{-21}\,\rm GeV^{-1}$ used above as a numerical input to the Route-2 amplitude estimate. R4 does not close by amplitude suppression: the same coupling that reproduces $\beta_{\rm obs}$ can, with a free normalization, also be tuned to reproduce ρ_Λ, but minimal ECH does not derive either the required ultralight mass $m_\theta\sim H_0$ or the fitted α/M from first principles—so the channel closes as a naturalness/explanatory-deficit objection (it relocates rather than solves the cosmological-constant problem) rather than as an amplitude exclusion. Its full derivation, and the companion consistency check against a spectator-ALP birefringence benchmark, are outside the present survey's scope [1, 12].

Within the four-channel enumeration of Sec. II, Routes R1–R4 cover the four parity-odd/dark-energy channels of this framework. Two operators occasionally raised as omissions from this four-channel enumeration — the

Jackiw–Pi gravitational Chern–Simons term $R\wedge\widetilde{R}$ and the parity-odd four-fermion partner of R1 carrying the $\gamma_{\rm BI}/(\gamma_{\rm BI}^2{+}1)\cdot 8\pi G$ coefficient — are closed in the companion analysis [1] (total derivative for constant coupling and R4-class otherwise, for the Jackiw–Pi term; Planck suppression plus vanishing mean field, inheriting R1, for the four-fermion partner); the dimension-6 parity-odd four-fermion basis is Fierz-closed and uniformly Planck-suppressed by the projection lemma of Appendix B, so only a non-minimal completion lies outside the established no-go. Within the four enumerated channels: R1 (NJL contact) is amplitude-suppressed by $M_{\rm Pl}^{-2}$ and parity-even. R2 (one-loop graviton corrections) is amplitude-suppressed by $M_{\rm Pl}^{-1}$ and the one-loop factor $\alpha_{\rm em}/(4\pi)$. R3 (Immirzi running) is mass-dimension-locked and additionally suppressed by the chiral-asymmetry beta function. R4 (parity-odd CMB coupling) is closed by a naturalness objection: with α/M treated as a free parameter, a spectator-ALP fit reproduces both $\beta_{\rm obs}$ and ρ_Λ, but minimal ECH does not derive $m_\theta\sim H_0$ or the fitted α/M, so the channel closes at the level of an explanatory deficit rather than an amplitude exclusion.

Tiered closure structure: R1–R3 achieve *mathematical amplitude closure* (the predicted amplitude is suppressed by explicit $M_{\rm Pl}$ powers and loop factors, placing it many orders below any observable threshold regardless of normalisation assumptions); R4 achieves *naturalness/explanatory closure* (the amplitude can be matched to observations but only at the cost of importing the cosmological-constant fine-tuning the channel was meant to resolve). These are logically distinct closure modes and are classified separately in the evidentiary table below.

a. Evidentiary status of each leg. So that the strength of the closure is not overread, we state explicitly the evidentiary level at which each route and each structural result is established. We use a three-tier scale: **(I) rigorous result** — a deductive consequence of stated equations/identities, holding exactly within an explicitly bounded scope; **(II) structural argument** — a qualitative or order-of-magnitude argument from established physics (symmetry, parity, naturalness) that does not turn on a fitted number; and **(III) ansatz-level dimensional estimate** — an amplitude budget evaluated under an explicitly-labeled scaling ansatz, honest to a factor but not a first-principles derivation. Table II classifies every leg on this scale. No leg is claimed at a level higher than this table records: in particular, the only Tier-I (rigorous) leg is the perturbation-transparency result for canonical scalar matter; R2–R3 are Tier-III ansatz-level estimates; and the R4 closure is a Tier-II *naturalness/explanatory-deficit* objection, *not* an amplitude exclusion and conditional on the on-shell scaling ansatz of Appendix A. The aggregate "four-route channel-level closure" is therefore exactly that: a channel-level (not operator-level) statement whose individual legs sit at the levels tabulated, reinforced but not promoted by the mechanism-class catalog of Sec. III.

The phenomenological parameter α/M introduced above is therefore best understood as the R4-bounded coupling required to match $\beta_{\rm obs}$, with the dark-energy density supplied by an unrelated sector (e.g. a quintessence field [13], a quintom-class two-field scenario [14], or by a positive cosmological constant), rather than by ECH-internal physics. This inversion of the prior mass-coupling lock is a core structural finding reused throughout this catalog.

V. THE OPERATOR-BASIS-COMPLETENESS ARGUMENT

The barrier catalog and the Route-2/Route-3 closures above bound the amplitude of *one* representative parity-odd operator. A referee may reasonably ask whether the same conclusion survives once the full local operator basis is considered. This section answers that question: we enumerate every local, gauge-invariant, diffeomorphism-covariant parity-odd density of mass dimension exactly four admitted by the minimal-ECH field content and show that the single-scale dimensional bound applies to the entire basis, not to one representative.

As a phenomenological ansatz for a direct parity-odd curvature coupling — motivated by, but not literally derived from, the Holst-plus-non-minimal-fermion construction of Mercuri [5] — consider the parity-odd term

$$S_{\rm eff}=\frac{\alpha}{M}\int e_I\wedge e_J\wedge\mathcal{F}^{IJ}[K,\mathring{R}],\qquad(5)$$

where $M=M_{\text{area-gap}}\sim M_{\rm Pl}/\sqrt{\gamma}$ is the LQG area-gap mass scale (from the LQG area-gap $\Delta\propto\gamma\,\ell_P^2$, the inverse-length / mass scale is $M_\Delta\sim M_{\rm Pl}/\sqrt{\gamma}$ up to numerical constants), α is a dimensionless coupling, and $\mathcal{F}^{IJ}[K,\mathring{R}]$ denotes the curvature two-form of the full (torsionful) connection, written as a functional of the contorsion K and the Levi-Civita curvature $\mathring{R}$; the component form Eq. (6) displays its leading contribution.

In components, the leading contribution reduces to

$$S_{\rm eff}=\int d^4x\,\sqrt{-g}\,\frac{\alpha}{M}\,\varepsilon^{\mu\nu\rho\sigma}\,e^I_\mu\,e^J_\nu\,\mathcal{F}_{IJ\rho\sigma},\qquad(6)$$

which has naive mass dimension $[\mathcal{L}_{\rm odd}]=+1$—three units short of the required $+4$ (with $[\alpha]=0$, $[M]=+1$, $[e^I_\mu]=0$, and $[\mathcal{F}_{IJ\rho\sigma}]=+2$, the integrand carries mass dimension $-1{+}2=+1$). We state this off-shell mismatch deliberately, and we address it immediately by identifying the physical foundation before using the shorthand.

Physical foundation: the dimension-4 off-shell operator basis.—The formally dimension-4, gauge-invariant, diffeomorphism-covariant off-shell operator basis is the closed local expansion

$$S^{(4)}_{\rm eff}=\int d^4x\,\sqrt{-g}\sum_n c_n\,\mathcal{O}^{[4]}_n,\qquad\{\mathcal{O}^{[4]}_n\}=\{\text{O1–O6}\},\qquad(7)$$

TABLE II. Evidentiary status of each closure leg, on the three-tier scale of the text: (I) rigorous result within stated scope, (II) structural argument, (III) ansatz-level dimensional estimate. The table records the highest level at which each leg is claimed; no leg is asserted more strongly elsewhere. "Rules out" is to be read at channel-amplitude granularity, not as an operator-level theorem.

Leg	**What it rules out (channel-amplitude level)**	**Evidentiary status**
Perturbation transparency (companion paper [1])	The Holst sector / Barbero–Immirzi parameter contributing to scalar or tensor perturbation observables, for canonical scalar matter	**(I) Rigorous** within stated scope: $T=0$ follows from zero scalar spin density, and the Holst dual contraction vanishes by the algebraic Bianchi identity. Excludes propagating-torsion, fermion-loop, dynamical-Immirzi, non-minimal-matter sectors.
R1 (NJL four-fermion contact)	The torsion-induced contact term sourcing coherent $w\!=\!-1$ dark energy	**(II)+(III)**: parity-even structure ($\langle J^5\rangle\!\approx\!0$, no coherent mean field) is a Tier-II algebraic fact; the $M_{\rm Pl}^{-2}$ amplitude suppression ($\sim$70 orders below ρ_Λ) is a Tier-III order-of-magnitude estimate from a standard Cartan torsion-elimination derivation.
R2 (one-loop graviton corrections)	One-loop promotion of $\gamma_{\rm BI}$ reaching the observed birefringence amplitude	**(III) Ansatz-level**: amplitude budget suppressed by $H_0/M_{\rm Pl}$ and $\alpha_{\rm em}/(4\pi)$ under an explicitly-labeled scaling ansatz; exploratory, not load-bearing.
R3 (Immirzi running)	Quantum running of $\gamma_{\rm BI}$ generating a dark-energy-relevant shift	**(II)+(III)**: mass-dimension lock is structural; the chiral-asymmetry beta-function bound (Benedetti–Speziale) is a Tier-III order-of-magnitude upper bound, deliberately loose and non-load-bearing ($\gtrsim 60$ orders of margin).
R4 (parity-odd CMB / spectator-ALP coupling)	Minimal ECH *deriving* the single coupling that yields both $\beta_{\rm obs}$ and ρ_Λ	**(II) Structural naturalness objection**, *not* an amplitude exclusion: a free-α/M ALP fit reproduces $\beta_{\rm obs}$, but ECH supplies neither $m_\theta\!\sim\!H_0$ nor the fitted α/M, relocating the CC problem. Conditional on the on-shell scaling ansatz (App. A).

where each $\mathcal{O}_n^{[4]}$ is a genuine dimension-4 Lagrangian density and each Wilson coefficient c_n is a dimensionless rational. The operators {O1–O6} constitute the *primary physical foundation* of the no-go: the single-scale NDA ceiling, the Fierz closure, and the Bianchi-vanishing results below are all derived properties of this off-shell dimension-4 basis; none depends on the on-shell shorthand Eq. (6) for its logical force. Because the underlying schematic invariants carry different naive dimensions under the paper's conventions ($[e]=0$, $[R]=+2$, physical torsion $[T]=[\kappa S]=+1$, spin current $[S]=[J^5]=+3$, $[\kappa]=[M_{\rm Pl}^{-2}]=-2$), the operator *definitions* include the explicit compensating $M_{\rm Pl}$ power that renders each density dimension-4; we do *not* attach a dimensionless c_n to a lower-dimension invariant. *Shorthand and its relation to the basis.*—Eq. (6) with its dimension-+1 integrand is the on-shell reduction of the dimension-4 basis after the algebraic Bianchi identity strips one curvature factor; it is a convenient dimensional-analysis shorthand, *not* the fundamental action term and *not* load-bearing for the no-go. To clarify the mass-dimension bookkeeping: the off-shell operators O1–O6 are genuinely dimension +4 (each carrying the compensating $M_{\rm Pl}^2$ factor that makes the Wilson coefficient dimensionless), while Eq. (6) represents the same physics on the torsion-free branch after the Bianchi identity reduces $[\mathcal{F}_{IJ\rho\sigma}]$ from +2 to +1; the apparent dimension-+1 deficit is an artifact of that on-shell reduction, not of any ill-defined operator. The Case II on-shell curvature dressing (inserting $R\sim M_{\rm Pl}^2$ to close the dimension count) is an *illustrative heuristic*, subordinate to the operator basis and used solely for dimensional-analysis aesthetics; it carries no independent logical weight in the no-go argument. Concretely, the six dimension-4 densities are (names as in Table III)

$$\begin{aligned} \mathcal{O}_1^{[4]} &= M_{\rm Pl}^2\,\varepsilon\, e^I e^J R_{IJ}, & \mathcal{O}_2^{[4]} &= M_{\rm Pl}^2\,\mathrm{NY}, \\ \mathcal{O}_3^{[4]} &= R^{IJ}\wedge R_{IJ}, & \mathcal{O}_4^{[4]} &= M_{\rm Pl}^2\,\varepsilon_{IJKL}T^{IJ}T^{KL}, \\ \mathcal{O}_5^{[4]} &= \varepsilon\, T\, e\, J^5, & \mathcal{O}_6^{[4]} &= M_{\rm Pl}^2\,\varepsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}, \end{aligned} \tag{8}$$

so that Pontryagin (O3, $[R\wedge R]=+4$) and axial–torsion

(O5, $[TeJ^5] = +1 + 0 + 3 = +4$) already sit at dimension 4 with a bare dimensionless coefficient, while the Holst dual, Nieh–Yan, torsion-square, and single-curvature densities (each of naive dimension +2 as component densities) are promoted to dimension 4 by the single available scale $M_{\rm Pl}^2$ inside the operator definition. This bookkeeping is essential: labelling the bare invariants as "dimension-4 with dimensionless c_n" would misstate their mass dimension, whereas the definitions above carry the $M_{\rm Pl}^2$ prefactors explicitly so that every $c_n \mathcal{O}_n^{[4]}$ is a bona-fide dimension-4 density. The physics conclusion is unchanged: the algebraic Cartan constraint $T = \kappa S$, the first (algebraic) Bianchi identity, and Chern–Weil exactness collapse this set to (i) topological total derivatives (O2, O3), (ii) the $M_{\rm Pl}^{-2}$-suppressed Fierz-closed four-fermion sector (O4, O5), or (iii) Bianchi-vanishing single-curvature densities (O1, O6).

Main-text completeness argument.—The collapse is closed by one finite enumeration plus two tensor identities, both verified symbolically in the released script `arxiv/scripts/dim4_parityodd_enumeration.py`. (a) The single-curvature parity-odd densities vanish identically: for any Riemann tensor obeying the pair antisymmetries and the algebraic Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$, $\varepsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = 0$, killing both O1 and O6 on the torsion-free branch. (b) The torsion-square and axial–torsion densities reduce under $T = \kappa S$, using $S_{abc}S^{abc} = -\frac{3}{8}(J^5 \cdot J^5)$ (from $S^{abc} = \frac{1}{4}\varepsilon^{abcd} J^5_d$ and the Lorentzian $\varepsilon_{abcd}\varepsilon^{abce} = -3!\,\delta^e_d$, consistent with the companion paper's torsion-elimination normalization [1]), to the parity-odd four-fermion contact operator $\kappa^2(J^5 \cdot J^5)$, which the Fierz-by-Fierz projection lemma (Appendix B) shows closes onto the finite $\{SS, VV, AA, PP\}$ basis with *no* escape operator and *no* change of $M_{\rm Pl}^{-2}$ power. (c) O2 (Nieh–Yan) and O3 (Pontryagin) are exact total derivatives, contributing zero to the local equations of motion and zero to the vacuum energy. Hence every admissible local dimension-4 parity-odd density in minimal ECH is topological, Fierz-basis-reducible, or Bianchi-vanishing, and none produces a $(\mathrm{meV})^4$ vacuum energy without a new light scale $\mu \ll M_{\rm Pl}$ or a protected symmetry — either of which is itself the tuning to be explained. The concrete operator definitions, the symbolic checks, and the Fierz matrix are given in full in Appendices A 1–B (Table III). Passing to the on-shell branch (algebraic torsion eliminated, one curvature factor stripped by the Bianchi identity) reduces this closed dimension-4 set to the single dimension-+1 representative displayed in Eq. (6): the +1-vs-+4 counting is a property of that on-shell reduction, not of an ill-defined action. The identification $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$ is therefore governed by a *single-scale NDA dimensional no-go* that holds identically for the formal dimension-4 basis and its dimension-+1 shorthand (Appendix A 1 verifies the closure survives at genuine dimension 4 without the on-shell dressing): the three-unit deficit forces the missing powers to $M_{\rm Pl}^3$ (the only scale in minimal ECH), so the natural density is $\sim M_{\rm Pl}^4$, never $(\mathrm{meV})^4$ (Appendix A). This is a dimensional-impossibility argument, not a derived amplitude, and hence not circular; it is treated as such throughout.

VI. DISCUSSION: SCOPE AND BOUNDARIES

Consistent with the evidentiary classification of Sec. III and Table II, we state explicitly what this survey does and does not establish.

What is established. Thirteen distinct mechanism-class constraints, covering seven foundational classes and six observational branches, collectively close the four enumerated minimal-ECH dark-energy channels (R1–R4) under stated assumptions. Two of those channels (R2, R3) are closed here at the amplitude level with margins exceeding sixty orders of magnitude; the closures are robust to $\mathcal{O}(1)$–$\mathcal{O}(10^{10})$ rescalings of their ansatz-level normalizations. The operator-basis-completeness argument (Sec. V) extends the governing dimensional bound from one representative operator to the complete local dimension-four parity-odd operator basis admitted by the minimal-coupling field content, with no escape operator identified within that basis.

What is not established. This is a *channel-level*, not a fully operator-level, no-go over the entire space of diffeomorphism-invariant gravitational operators: the completeness argument is scoped to local, gauge-invariant, mass-dimension-exactly-four densities built from the minimal-ECH field content (tetrad, algebraically-fixed torsion, Holst term, minimally/totally-antisymmetrically coupled fermions). It does not enumerate derivative four-fermion terms suppressed by additional momentum powers, curvature-torsion mixed invariants of higher order, multi-species chiral structures beyond the single-species minimal content, dynamical-Immirzi-field completions, or non-minimal (trace/tensor) torsion irreps — any of these constitutes a genuine escape from the minimal-coupling scope, not a loophole within it. A non-minimal completion introducing a new light scale $\mu \ll M_{\rm Pl}$, or a protected symmetry zeroing the Planck-scale piece, would itself evade the single-scale bound — but such a scale or symmetry *is* the fine-tuning the mechanism was meant to explain, so evading the no-go this way relocates rather than solves the cosmological-constant problem.

The barrier catalog is a map of failure modes of mixed evidentiary strength, not thirteen independently decisive theorems: five entries (B5, B6, B7, B10, B13) are general naturalness or classification arguments rather than ECH-specific calculations, and one (B9) is an explicitly heuristic closure conditional on stated equilibrium assumptions that dissipative or particle-producing bounces can violate. Only the perturbation-transparency result (B14, established in the companion paper) and the Route-1 torsion-elimination derivation (also companion-paper content) are Tier-I rigorous results within their stated scope. The R2/R3 amplitude estimates are Tier-

III ansatz-level bounds: honest to a factor, not first-principles derivations, though robust to their own ansatz uncertainty by tens of orders of magnitude. The R4 closure is a Tier-II naturalness/explanatory-deficit objection rather than an amplitude exclusion, and its numerical inputs (α/M, $\beta_{\rm obs}$) are used here only as context for the Route-2 estimate; its own derivation is a companion-paper (or literature) result, not reproduced in this survey.

Finally, every barrier and closure in this catalog is specific to the *minimal* ECH mechanism class. Other bouncing cosmologies — notably quintom scenarios with independent dynamical fields, or bounces with non-minimal torsion couplings — access observational and structural channels outside this catalog's scope and are not addressed by any result in this paper.

VII. CONCLUSIONS

We have presented a structural no-go survey of minimal Einstein–Cartan–Holst spin-torsion routes to late-time dark energy and bounce phenomenology. Through systematic analysis of 7 foundation classes and 6 observational branches, we catalog 13 distinct mechanism-class constraints (14 historical entries, B8 subsumed by B14) that map the minimal-ECH parameter space and identify which dark-energy pathways require phenomenological assumption or physics beyond the minimal framework. Two of the four enumerated dark-energy channels, Route 2 (one-loop graviton-sector corrections to the Holst term) and Route 3 (quantum running of the Barbero–Immirzi parameter), are closed here at the amplitude level, one one-loop-grounded and one from a derived integrated renormalization-group flow, each leaving more than sixty orders of magnitude of suppression margin. An operator-basis-completeness argument extends the governing single-scale dimensional bound from one representative parity-odd operator to the complete local dimension-four operator basis admitted by the minimal-coupling field content: every member of that basis is topological, Fierz-basis-reducible, or Bianchi-vanishing, with no identified escape operator within the stated minimal scope.

The catalog's collective force is deliberately not overstated: it is presented with an explicit evidentiary classification distinguishing rigorous results, structural arguments, and ansatz-level estimates, and its scope is bounded to the minimal-coupling ECH field content. Non-minimal completions — an added light scale or non-minimal torsion coupling — lie explicitly outside this survey and would themselves reintroduce the fine-tuning the closed channels were meant to avoid. Read together with the companion paper's torsion-elimination and perturbation-transparency results [1], this survey completes the channel-level accounting of minimal-ECH dark-energy routes: all four enumerated channels close, by amplitude suppression (R1–R3) or by naturalness/explanatory deficit (R4), under the stated minimal-coupling assumptions.

Data and Code Availability

The symbolic verification scripts underlying the operator-basis-completeness argument (Sec. V, Appendices A–B) are publicly available in the same repository as the companion papers:

`arxiv/scripts/dim4_parityodd_enumeration.py`
`arxiv/scripts/fierz_lemma_check.py`

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Appendix A: Dimensional Status of the Parity-Odd Operator

The parity-odd operator (Eq. 6) has off-shell mass dimension +1, not the +4 required for a local Lagrangian density:

$$[\alpha/M] = -1, \quad [\varepsilon^{\mu\nu\rho\sigma} e^I_\mu e^J_\nu \mathcal{F}_{IJ\rho\sigma}] = +2 \implies [\mathcal{L}_{\rm odd}] = +1. \tag{A1}$$

Rather than an obstacle to be bridged by assumption, this three-unit deficit *is* the physical content of the no-go, once read through effective-field-theory power counting. A relevant (dimension $d < 4$) operator carries, by naive dimensional analysis, a Wilson coefficient of size Λ^{4-d} set by the EFT cutoff [15–17]. In minimal ECH the *only* available scale is $\Lambda \sim M_{\rm Pl}$: there is no intermediate threshold and no assumed cancellation. The three missing mass powers are therefore *forced* to be $M_{\rm Pl}^3$ — one cannot conjure a light scale from a theory that has none — and there are exactly two dimensionally-consistent ways to supply them.

Case I — coefficient dressing (local NDA reading).— The natural coefficient of the relevant operator is $c \sim M_{\rm Pl}^{4-1} = M_{\rm Pl}^3$, and the only static (VEV) piece of $\varepsilon\, e\, e\, \mathcal{F}$ that can act as a cosmological constant is the dimension-$+1$ torsion/topological density T, giving $\rho \sim M_{\rm Pl}^3 \langle T\rangle$. A coherent cosmological torsion is forbidden today by parity and isotropy ($\langle T\rangle \to 0$, hence $\rho \to 0$), while at the bounce $\langle T\rangle \sim M_{\rm Pl}$ returns $\rho \sim M_{\rm Pl}^4$. Neither reproduces $(\mathrm{meV})^4$.

Case II — on-shell curvature dressing.—Retaining the small coefficient α/M (dimension -1) and inserting on-shell bounce curvature $R \sim M_{\rm Pl}^2$ gives

$$\rho_\Lambda^{\rm bounce} \sim (\alpha/M)\, M_{\rm Pl}^5 \sim 10^{-2}\, M_{\rm Pl}^4, \tag{A2}$$

a bounce-era density that must dilute by $e^{-3N_{\rm tot}} \sim 10^{-122}$ ($N_{\rm tot} \approx 94$) to reach $\rho_\Lambda^{\rm obs}$. The $+1 \to +4$ mass-dimension promotion in Case II is a *heuristic dimensional argument*: absorbing the three missing mass powers into on-shell bounce curvature factors ($R \sim M_{\rm Pl}^2$) is a dimensional assignment consistent with the single-scale power-counting logic, but it does not constitute a full field-theoretic formalization (e.g., a controlled effective-operator construction demonstrating that the specific curvature insertion is gauge-invariant, diffeomorphism-covariant, and free of operator-mixing contributions at the bounce scale). This limitation is explicitly disclosed; the NDA order-of-magnitude conclusion that the natural density sits at $\rho_\Lambda^{\rm ECH} \sim M_{\rm Pl}^4$ is unchanged by this caveat, since both Case I and Case II land at the same Planck-scale ceiling via the single-scale power-counting logic, regardless of which off-shell completion is adopted.

The single-scale NDA no-go.—In both admissible completions the three missing mass powers are $M_{\rm Pl}$ (the only scale), so any dimensionally-consistent minimal-ECH parity-odd source sits at

$$\rho_\Lambda^{\rm ECH} \sim M_{\rm Pl}^4 \quad (\text{up to } \mathcal{O}(1)\text{–}\mathcal{O}(10^{-2})), \qquad \textit{never}\,(\mathrm{meV})^4. \tag{A3}$$

Torsion cannot *be* dark energy because dimensional analysis pins its natural density at $M_{\rm Pl}^4$ and minimal ECH supplies no light scale (no $m_\theta \sim H_0$, no dynamically-small α/M) to bridge the resulting ~ 122-order hierarchy: the cosmological-constant problem re-appears untouched. This is a single-scale power-counting bound, not a fitted amplitude. Because *no* positive ρ_Λ is derived (and none is needed), the closure is a naturalness / amplitude-ceiling no-go and *cannot be circular*: a charge of circularity requires a claimed derivation of the conclusion from an assumption of that same conclusion, and here there is no derived amplitude at all — the $+1 \to +4$ dimensional gap, read correctly, is itself the mechanism.

Residual assumption (explicit, not fabricated).—The bound assumes single-scale EFT: $\Lambda \sim M_{\rm Pl}$, no intermediate threshold $\mu \ll M_{\rm Pl}$, and no exact symmetry/topological cancellation. A non-minimal UV completion introducing a new light scale $\mu \ll M_{\rm Pl}$ in the coefficient ($c \sim \mu^a M_{\rm Pl}^{3-a}$), or a symmetry that zeroes the $M_{\rm Pl}^4$ piece and leaves a protected small remainder, could evade the estimate — but such a scale or symmetry *is* the tuning the mechanism was meant to explain, relocating rather than solving the cosmological-constant problem. Minimal ECH contains neither, which is precisely why the no-go holds *within minimal ECH*. Full closure would require a matching calculation over parity-odd ECH-compatible completions showing that any such completion either (i) reproduces the $M_{\rm Pl}^4$ NDA estimate, or (ii) requires an added light scale/symmetry that is itself the tuning being explained; that matching calculation is left to a companion treatment. For the minimal-ECH field content this matching is in fact immediate rather than deferred: the finite operator basis fixed by the algebraic Cartan constraint and the totally-antisymmetric minimal spin current (Sec. II) contains no operator of lower off-shell dimension than the $+1$ operator bounded above, so single-scale NDA applied to that ceiling bounds every member of the basis; only a non-minimal light scale or protected cancellation—case (ii)—evades it, and that is the tuning being explained.

Inflationary dilution ($\mathcal{D}_{\rm inf} \sim e^{-3N_{\rm tot}}$, with $\mathcal{D}_{\rm inf} \equiv e^{-3N_{\rm tot}}$) then yields $\rho_\Lambda = \Xi\, M_{\rm Pl}^4$ with $\Xi = (\alpha/M)\, M_{\rm Pl}\, \mathcal{D}_{\rm inf}$ under Eq. (A2). The genuine cosmological-constant hierarchy is $M_{\rm Pl}^4/\rho_\Lambda^{\rm obs} \sim 10^{19\,\mathrm{GeV}\times 4}/(10^{-3}\,\mathrm{eV})^4 \sim 10^{122}$, i.e. ~ 120 orders of magnitude (the bounce-scale density entering this hierarchy is $\rho_{\rm bounce} \sim M_{\rm Pl}^4$, not the local pseudo-density $\rho_\Lambda^{\rm bounce} \sim 10^{-2}\, M_{\rm Pl}^4$ that Eq. (A2) labels). The dilution factor required to bridge $\sim 10^{122}$ from $M_{\rm Pl}^4$ down to the observed ρ_Λ is $\mathcal{D}_{\rm inf} \sim e^{-3N_{\rm tot}} \sim 10^{-122}$, giving $N_{\rm tot} \approx 122 \ln 10/3 \approx 94$ e-folds.

Sharper dependency statement.—The precise value of $N_{\rm tot}$ *does* depend on the ansatz choice in Eq. (A2); however, the *overall scale separation* between $\rho_\Lambda^{\rm bounce}$ and $\rho_\Lambda^{\rm obs}$ is $\sim$120 orders of magnitude under *any* non-cancellation assumption (whether the bounce-scale density is $M_{\rm Pl}^4$, $10^{-2}\, M_{\rm Pl}^4$, or $10^{-4}\, M_{\rm Pl}^4$ shifts the required e-fold count by $\mathcal{O}$(a few), not by orders of magnitude), and this scale separation is what drives the 13 mechanism-class structural barriers (Sec. III). The barriers close the bounce-to-dark-energy route on amplitude-budget and operator-counting grounds whose qualitative conclusions are ansatz-independent; only the precision of the headline $N_{\rm tot}$ figure is ansatz-dependent. Readers should therefore treat $N_{\rm tot} \approx 92$–94 as an order-of-magnitude estimate, not as a precise number; the broader structural closure does not depend on the precise e-fold value.

The on-shell scaling of Case II is labeled explicitly as a phenomenological ansatz rather than as a derivation. The controlled EFT-level construction it stands in for — an *operator-basis closure* for the genuine local dimension-$+4$ parity-odd sector of minimal ECH — is, however, not deferred: it is supplied in the next subsection, where we enumerate that basis and show that single-scale NDA closure survives without the Case II dressing.

1. Genuine Dimension-Four Parity-Odd Completion: the Operator Basis Survives Single-Scale Closure

The dimension-+1 operator of Eq. (6) is a presentationally-simplified representative. A referee may reasonably ask for the *fully gauge-invariant, diffeomorphism-covariant, local dimension*-4 completion directly. We therefore enumerate every local, Lorentz- and diffeomorphism-invariant, parity-odd density of mass dimension exactly +4 that the minimal-ECH field content admits — tetrad e^I_μ, Holst term, *algebraic* torsion fixed by the Cartan constraint $T^{abc} = \kappa S^{abc}$ ($\kappa = M_{\rm Pl}^{-2}$), and minimally-coupled canonical matter — and determine the fate of each. No new light scale $\mu \ll M_{\rm Pl}$, no dynamical Immirzi field, no propagating torsion, and no non-minimal (trace/tensor) torsion irreps are admitted, as these lie outside the stated minimal scope. Building-block dimensions: $[e] = 0$, $[\omega] = +1$, $[R] = +2$, $[S] = +3$ (Dirac bilinear), so via Cartan each torsion factor carries $[\kappa S] = +1$. Under these assignments the bare invariants $\varepsilon\, eeR$, Nieh–Yan, and T^2 are component densities of naive dimension +2, while Pontryagin $R \wedge R$ and the axial–torsion current structure $\varepsilon\, TeJ^5$ are already dimension +4. To write a genuine dimension-4 Lagrangian with *dimensionless* Wilson coefficients we therefore include the single available scale $M_{\rm Pl}^2$ explicitly in each operator definition [Eq. (8)], rather than attaching a dimensionless c_n to a lower-dimension object. Column "dim" below records the naive dimension of the *bare* schematic invariant; the tabulated operator $\mathcal{O}_n^{[4]}$ is that invariant multiplied by the prefactor of column "prefactor" so that $[\mathcal{O}_n^{[4]}] = +4$ throughout. The "Fate" column records the reduction of the *bare* invariant (hence its κ-power tracks the bare dimension: O4's bare torsion-square $\varepsilon TT \to \kappa^2(J^5 \cdot J^5)$ at dim +2, O5's bare $\varepsilon TeJ^5 \to \kappa(J^5 \cdot J^5)$ at dim +4). Restoring each prefactor, the two *genuine* dimension-4 densities land on the same member of the Fierz basis at the same power: $\mathcal{O}_4^{[4]} = M_{\rm Pl}^2 \cdot \kappa^2(J^5 \cdot J^5) = \kappa(J^5 \cdot J^5)$ [using $M_{\rm Pl}^2\kappa^2 = \kappa$] $= \mathcal{O}_5^{[4]}$, both at dimension +4 and both $M_{\rm Pl}^{-2}$-suppressed — consistent with the "+4 throughout" contract.

Two tensor identities carry the enumeration; both are verified symbolically in the released script `arxiv/scripts/dim4_parityodd_enumeration.py`.

Check A — single-curvature parity-odd invariants vanish.—For any Riemann tensor obeying the pair anti-symmetries and the first (algebraic) Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$,

$$\varepsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = 0 \quad \text{(identically)}, \tag{A4}$$

verified by imposing the identity as linear constraints on a generic symbolic curvature and contracting. This is the same Bianchi-vanishing that drives the perturbation-transparency result established in the companion paper's zero-spin-branch analysis [1]; it kills *both* O1 (the Holst dual) and O6 (any single-curvature parity-odd density) on the torsion-free branch. The torsionful piece of R is $\mathcal{O}(\kappa S)$ and merely feeds the O4/O5 four-fermion channel.

Check D — torsion-square collapse.—The minimal (totally antisymmetric) spin current is dual to the axial vector, $S^{abc} = \frac{1}{4}\varepsilon^{abcd}\bar{\psi}\gamma_d\gamma^5\psi = \frac{1}{4}\varepsilon^{abcd}J^5_d$; using the Lorentzian (mostly-plus, $\varepsilon^{0123} = +1$) contraction $\varepsilon_{abcd}\varepsilon^{abce} = -3!\,\delta^e_d$ (verified symbolically) gives $S_{abc}S^{abc} = \frac{1}{16}\varepsilon_{abcd}\varepsilon^{abce}J^{5d}J^5_e = -\frac{3}{8}(J^5 \cdot J^5)$, consistent with the companion paper's torsion-elimination normalization [1]. Every ε-contracted torsion-square (O4) therefore reduces under $T = \kappa S$ to $\kappa^2(J^5 \cdot J^5)$, and the linear axial–torsion coupling (O5) to $\kappa\,(J^5 \cdot J^5)$ — both members of the Fierz-closed basis (App. B), bounded uniformly at $M_{\rm Pl}^{-2}$ with natural coefficient $\sim M_{\rm Pl}^4$ by single-scale NDA.

Structural entries.—O2 (Nieh–Yan) is an exact 4-form, $d(e_I \wedge T^I) = T_I \wedge T^I - e_I \wedge e_J \wedge R^{IJ}$ [18], and O3 (Pontryagin) is exact by Chern–Weil; both are total derivatives contributing zero to the local equations of motion and zero to the vacuum energy, independently of torsion.

Verdict.—Every admissible local dimension-4 parity-odd density in minimal ECH *within the enumerated set {O1–O6} at the stated power-counting order* (the non-enumerated classes of the Fierz scope caveat below — derivative four-fermion terms, higher-curvature–torsion mixed invariants, multi-species chiral structures, dynamical-Immirzi coefficients, non-minimal torsion irreps — are explicitly excluded) is (i) a topological total-derivative term (O2, O3) contributing nothing to the EOM or vacuum energy; (ii) a four-fermion contact term (O4, O5) that collapses via the algebraic Cartan constraint to the Fierz-closed basis with natural coefficient $\sim M_{\rm Pl}^4$; or (iii) a single-curvature parity-odd density (O1, O6) that vanishes by Eq. (A4). No genuine local dimension-4 parity-odd operator in minimal ECH produces a $(\text{meV})^4$ vacuum energy without introducing a new light scale $\mu \ll M_{\rm Pl}$ or a protected symmetry — and either *is* the tuning the mechanism was meant to explain. **Single-scale NDA closure therefore survives at genuine dimension 4**, without recourse to the Case II on-shell dressing: the $+1 \to +4$ dressing is dispensable once the true dimension-4 basis is enumerated, because every element is topological, Fierz-basis-reducible, or Bianchi-vanishing. The dimension-+1 representative of Eq. (6) is a simplified stand-in for this closed set, not an ad-hoc object requiring an uncontrolled promotion.

Appendix B: Fierz-by-Fierz Projection Lemma for the Minimal-ECH Four-Fermion Basis

The torsion-elimination step (established in the companion paper's torsion-elimination analysis [1]) generates, at the single prefactor $\kappa = 8\pi G = M_{\rm Pl}^{-2}$, the axial–

TABLE III. Local dimension-4 parity-odd densities of minimal ECH, their coefficient dimensions, and their fate. The bare schematic invariant of column 3 has naive dimension "dim"; the genuine dimension-4 density $\mathcal{O}_n^{[4]}$ [Eq. (8)] is that invariant times the listed prefactor, carried by a *dimensionless* Wilson coefficient c_n. All entries are topological total derivatives ($\to$ zero equations of motion), reduce under the algebraic Cartan constraint to the Fierz-closed four-fermion basis (App. B, natural coefficient $\sim M_{\rm Pl}^4$ by single-scale NDA), or vanish by the first (algebraic) Bianchi identity. None yields a $(\mathrm{meV})^4$ vacuum energy without a new light scale.

#	dim (bare)	Bare invariant (schematic)	Prefactor	Fate
O1	2	$\varepsilon\, e^I e^J R_{IJ}$ (Holst dual)	$M_{\rm Pl}^2$	vanishes (Bianchi, Check A)
O2	2	$\mathrm{NY} = d(e_I \wedge T^I)$ (Nieh–Yan)	$M_{\rm Pl}^2$	exact 4-form $\to$ 0 EOM
O3	4	$R^{IJ} \wedge R_{IJ}$ (Pontryagin)	1	exact $\to$ 0 EOM
O4	2	$\varepsilon_{IJKL} T^{IJ} T^{KL}$ (torsion^2)	$M_{\rm Pl}^2$	$\to \kappa^2 (J^5 \cdot J^5)$, Fierz basis
O5	4	$\varepsilon\, T\, e\, J^5$ (axial–torsion)	1	$\to \kappa (J^5 \cdot J^5)$, Fierz basis
O6	2	$\varepsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$	$M_{\rm Pl}^2$	vanishes (Bianchi, Check A)

axial contact term $c_{AA}\,(J^5 \cdot J^5)$ (the only structure present under *minimal* fermion coupling, per Freidel–Minic–Takeuchi [4]) and—only once a *non-minimal* fermion coupling is admitted—the vector–axial Holst partner $c_{VA}\,(J \cdot J^5)$. We prove that the complete Fierz rearrangement of these operators closes onto the enumerated dimension-≤ 6 parity-relevant basis with *no* escape operator and *no* change of $M_{\rm Pl}$ power.

On the 16-dimensional Clifford basis, grouped into the five Lorentz classes $(S, V, T, A, P) = (\mathbf{1}, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu\gamma^5, \gamma^5)$, Fierz rearrangement of a current–current product is the linear map $F_{AB} = \frac{1}{4}\,\mathrm{Tr}_{\rm class}(\Gamma^A \Gamma_B \Gamma_A \Gamma^B)/d_B$, $d_B = \mathrm{Tr}(\Gamma^B \Gamma_B)$. Computed from the explicit Dirac matrices in the paper's mostly-plus signature (script `arxiv/scripts/fierz_lemma_check.py`), it equals the standard matrix [19, 20]

$$F = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 4 & -2 & 0 & 2 & -4 \\ 6 & 0 & -2 & 0 & 6 \\ 4 & 2 & 0 & -2 & -4 \\ 1 & -1 & 1 & -1 & 1 \end{pmatrix}, \qquad F^2 = \mathbb{1}, \tag{B1}$$

in the order (S, V, T, A, P); the involution $F^2 = \mathbb{1}$ is verified symbolically. Applying F to the generated operators gives the exact decompositions

$$(J^5 \cdot J^5) \xrightarrow{\text{Fierz}} \tfrac{1}{4}SS + \tfrac{1}{2}VV - \tfrac{1}{2}AA - \tfrac{1}{4}PP, \tag{B2}$$

$$(J \cdot J) \xrightarrow{\text{Fierz}} \tfrac{1}{4}SS - \tfrac{1}{2}VV + \tfrac{1}{2}AA - \tfrac{1}{4}PP, \tag{B3}$$

while the parity-odd $(J \cdot J^5)$ Holst partner rotates only within the $\{V, A\}$ block ($F_{VA} = F_{AV} = \frac{1}{2}$), remaining a dimension-6 parity-odd cross term. Every produced structure lies in the closed set $\{SS, PP, VV, AA\}$ (with $\{V, A\}$ for the cross term); the tensor channel does not appear, and no operator escapes the enumerated basis. Because every coefficient in F is a dimensionless rational, the Fierz map is an $O(1)$ recombination of the *same* four ψ fields and therefore preserves the $\kappa = M_{\rm Pl}^{-2}$ prefactor exactly. This is the explicit Fierz-by-Fierz projection lemma: within single-species minimal ECH the generated four-fermion tower is Fierz-closed and uniformly $M_{\rm Pl}^{-2}$-suppressed, so the single-scale NDA ceiling of App. A that bounds one representative bounds the entire finite basis. (For multiple fermion species the Lorentz-class closure and the $M_{\rm Pl}$-power argument are unchanged—Fierz acts on Dirac structure, not flavor—and only the flavor-off-diagonal coefficient bookkeeping is added.) The only operators evading this bound require a *non-minimal* completion (a new light scale $\mu \ll M_{\rm Pl}$, trace/tensor torsion irreps, or a dynamical Chern–Simons field), which is the stated scope boundary of the no-go.

Scope of the lemma (what it establishes and what it does not).—The Fierz lemma establishes basis-completeness of the *parity-odd four-fermion contact sector* at $M_{\rm Pl}$-power-counting within the minimal ECH field content (tetrad, algebraic torsion via Cartan constraint, Holst term, and minimally-coupled fermion). It does *not* enumerate: derivative four-fermion terms suppressed by additional powers of momentum/$M_{\rm Pl}$; curvature-torsion mixed invariants built from higher-order contractions of the Riemann tensor and spin current; chiral/flavor structures involving independent fermion species beyond the single-species minimal content; dynamical topological coefficients arising from a promoted Immirzi field; or non-minimal torsion irreps (trace-vector and tensor irreps) that appear only when the minimal coupling assumption is relaxed. The claim is scoped to the minimal-ECH field content at the stated power-counting order; it is not a complete operator-level no-go over all diffeomorphism-invariant operators in the full gravitational EFT space.

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